11 Verification of Orthonormality

It is instructive to verify one of the orthonormality relations by direct computation. Using the Cartesian expansions (34) we form the scalar product

$$\begin{aligned}\hat{r}\cdot\hat{r} &= (\sin\theta\cos\phi)^2 + (\sin\theta\sin\phi)^2 + (\cos\theta)^2 \\ &= \sin^2\theta(\cos^2\phi + \sin^2\phi) + \cos^2\theta \\ &= \sin^2\theta + \cos^2\theta = 1.\end{aligned}$$

A similar calculation yields $\hat{\theta}\cdot\hat{\theta} = 1$ and $\hat{\phi}\cdot\hat{\phi} = 1$. The mixed products vanish; for example

$$\begin{aligned}\hat{r}\cdot\hat{\theta} &= \sin\theta\cos\phi\cdot\cos\theta\cos\phi + \sin\theta\sin\phi\cdot\cos\theta\sin\phi + \cos\theta\cdot(-\sin\theta) \\ &= \sin\theta\cos\theta(\cos^2\phi + \sin^2\phi) - \sin\theta\cos\theta \\ &= 0.\end{aligned}$$

The same algebra recovers all of the cross-product identities listed in (29).

12 Infinitesimal Elements

12.1 Infinitesimal line element

An infinitesimal displacement receives independent contributions from each of the three spherical coordinates. A change dr produces a radial displacement $dr\,\hat{r}$. A change $d\theta$ produces an arc-length displacement $r\,d\theta$ along the meridian, hence the term $r\,d\theta\,\hat{\theta}$. A change $d\phi$ produces an arc-length displacement $r\sin\theta\,d\phi$ along the parallel of latitude, hence the term $r\sin\theta\,d\phi\,\hat{\phi}$. Collecting these contributions we obtain

$$d\vec{r} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}. \tag{40}$$

The magnitude of the displacement is therefore

$$|d\vec{r}| = \sqrt{(dr)^2 + (r\,d\theta)^2 + (r\sin\theta\,d\phi)^2}.$$

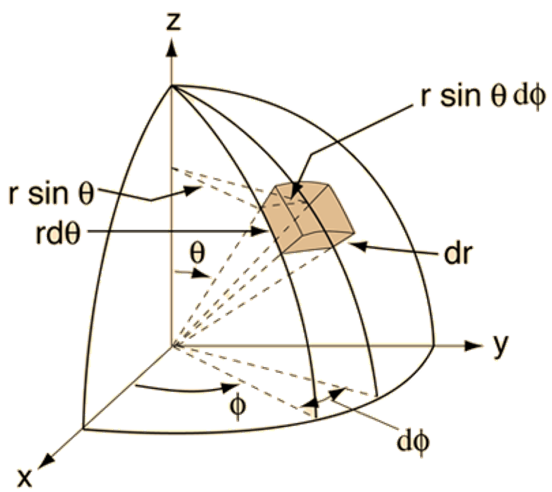

Figure 9: Infinitesimal displacements in spherical coordinates. The three mutually orthogonal increments are $dr\,\hat{r}$, $r\,d\theta\,\hat{\theta}$ and $r\sin\theta\,d\phi\,\hat{\phi}$.

12.2 Area elements on a sphere

On the surface of a sphere of fixed radius r the two angular increments generate a rectangular patch whose sides have lengths $r\,d\theta$ and $r\sin\theta\,d\phi$. The area of that patch is therefore

$$dA_r = r^2 \sin\theta\, d\theta\, d\phi. \tag{41}$$

When the surface is oriented outward the corresponding vector area element is

$$d\vec{A}_r = r^2 \sin\theta\, d\theta\, d\phi\, \hat{r}. \tag{42}$$

Similarly, other two vector area elements are

$$d\vec{A}_\theta = r \sin\theta\, dr\, d\phi\, \hat{\theta}. \tag{43}$$

and

$$d\vec{A}_\phi = r\, dr\, d\theta\, \hat{\phi}. \tag{44}$$

12.3 Volume element

The volume of an infinitesimal spherical shell is the product of the three edge lengths that appear in the line element:

$$dV = r^2 \sin\theta\, dr\, d\theta\, d\phi. \tag{45}$$

13 Summary of Key Formulae

Quantity	Spherical expression
Position vector	$\vec{r} = r\hat{r}$
Line element	$d\vec{r} = dr\,\hat{r} + r\,d\theta\,\hat{\theta} + r\sin\theta\,d\phi\,\hat{\phi}$
Area element (sphere)	$dA = r^2 \sin\theta\, d\theta\, d\phi$
Volume element	$dV = r^2 \sin\theta\, dr\, d\theta\, d\phi$
Cartesian $\leftrightarrow$ spherical	$x = r\sin\theta\cos\phi, \quad y = r\sin\theta\sin\phi, \quad z = r\cos\theta$

The orthonormality relations, the unit-vector transformations and the component-transformation matrices complete the basic tool required for calculations in spherical polar coordinates.

14 Newton's Second Law in Orthogonal Coordinate Systems

Newton's first law is theoretically important because it defines the class of inertial frames, yet it does not by itself determine the motion of a body. The third law is essential when one must identify the internal forces inside a multi-particle system. Once the forces that act on a given particle are known, however, the second law

$$\vec{F}(\vec{r}, \vec{v}, t) = m\vec{a} \tag{46}$$

becomes the working equation that determines the trajectory. Equation (46) is a second-order vector differential equation. The most practical way to solve it is to resolve every vector into components relative to a chosen orthogonal basis.

15 Cartesian Coordinates

The conceptually simplest basis is the fixed Cartesian triad $(\hat{\imath}, \hat{\jmath}, \hat{k})$. In that basis we write

$$\vec{F} = F_x\hat{\imath} + F_y\hat{\jmath} + F_z\hat{k}, \tag{47}$$

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}, \tag{48}$$

$$\vec{a} = \ddot{x}\hat{\imath} + \ddot{y}\hat{\jmath} + \ddot{z}\hat{k}. \tag{49}$$

Substitution into (46) immediately yields three independent scalar equations

$$\begin{aligned} F_x &= m\ddot{x}, \\ F_y &= m\ddot{y}, \\ F_z &= m\ddot{z}. \end{aligned} \tag{50}$$

Each equation is simply Newton's second law written for one Cartesian direction. Because the unit vectors are constant, no additional kinematic terms appear.

16 Plane Polar Coordinates

When the motion is confined to a plane and possesses axial symmetry, plane polar coordinates (r, θ) are far more convenient. The position vector is simply

$$\vec{r} = r\hat{r}, \tag{51}$$

where the radial unit vector admits the Cartesian expansion

$$\hat{r} = \cos\theta\,\hat{\imath} + \sin\theta\,\hat{\jmath}. \tag{52}$$

The orthogonal partner is

$$\hat{\theta} = -\sin\theta\,\hat{\imath} + \cos\theta\,\hat{\jmath}. \tag{53}$$

16.1 Velocity

Differentiate (51) with respect to time:

$$\vec{v} = \frac{d\vec{r}}{dt} = \dot{r}\hat{r} + r\frac{d\hat{r}}{dt}. \tag{54}$$

From (52) we obtain

$$\frac{d\hat{r}}{dt} = (-\sin\theta\,\dot{\theta})\hat{\imath} + (\cos\theta\,\dot{\theta})\hat{\jmath} = \dot{\theta}\,(-\sin\theta\,\hat{\imath} + \cos\theta\,\hat{\jmath}) = \dot{\theta}\,\hat{\theta}. \tag{55}$$

Consequently the velocity takes the orthogonal decomposition

$$\vec{v} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta}. \tag{56}$$

The first term is the radial speed; the second term is the tangential speed.

16.2 Acceleration

Differentiate (56) once more:

$$\vec{a} = \frac{d\vec{v}}{dt} = \ddot{r}\hat{r} + \dot{r}\frac{d\hat{r}}{dt} + \big(\dot{r}\dot{\theta} + r\ddot{\theta}\big)\hat{\theta} + r\dot{\theta}\frac{d\hat{\theta}}{dt}. \tag{57}$$

We already know $d\hat{r}/dt = \dot{\theta}\hat{\theta}$. Differentiating (53) yields

$$\frac{d\hat{\theta}}{dt} = (-\cos\theta\,\dot{\theta})\hat{\imath} + (-\sin\theta\,\dot{\theta})\hat{\jmath} = -\dot{\theta}\,(\cos\theta\,\hat{\imath} + \sin\theta\,\hat{\jmath}) = -\dot{\theta}\,\hat{r}. \tag{58}$$

Substitution of these two derivatives into (57) produces

$$\begin{aligned}\vec{a} &= \ddot{r}\hat{r} + \dot{r}(\dot{\theta}\hat{\theta}) + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + r\dot{\theta}(-\dot{\theta}\hat{r}) \\ &= (\ddot{r} - r\dot{\theta}^2)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}. \end{aligned} \tag{59}$$

Newton's second law in plane polar coordinates therefore reads

$$\begin{aligned} F_r &= m(\ddot{r} - r\dot{\theta}^2), \\ F_\theta &= m(r\ddot{\theta} + 2\dot{r}\dot{\theta}). \end{aligned} \tag{60}$$

16.3 Circular motion

When a particle is constrained to move on a circle of fixed radius $r = \text{const}$, one has $\dot{r} = \ddot{r} = 0$. Equations (60) reduce to

$$\begin{aligned} F_r &= -mr\dot{\theta}^2 = -m\frac{v^2}{r}, \\ F_\theta &= mr\ddot{\theta}. \end{aligned} \tag{61}$$

The radial component is the familiar centripetal force; the tangential component is responsible for any change in the angular speed.

17 Spherical Polar Coordinates

We now derive the analogous expressions in three-dimensional spherical coordinates (r, θ, ϕ). The position vector is again

$$\vec{r} = r\hat{r}, \tag{62}$$

but the radial unit vector now depends on both angular coordinates:

$$\hat{r} = \sin\theta\cos\phi\,\hat{\imath} + \sin\theta\sin\phi\,\hat{\jmath} + \cos\theta\,\hat{k}. \tag{63}$$

The remaining two unit vectors are

$$\hat{\theta} = \cos\theta\cos\phi\,\hat{\imath} + \cos\theta\sin\phi\,\hat{\jmath} - \sin\theta\,\hat{k}, \tag{64}$$

$$\hat{\phi} = -\sin\phi\,\hat{\imath} + \cos\phi\,\hat{\jmath}. \tag{65}$$

17.1 Time derivatives of the unit vectors

Differentiating (63)–(65) with respect to time yields the standard kinematic identities

$$\dot{\hat{r}} = \dot{\theta}\,\hat{\theta} + \sin\theta\,\dot{\phi}\,\hat{\phi}, \tag{66}$$

$$\dot{\hat{\theta}} = -\dot{\theta}\,\hat{r} + \cos\theta\,\dot{\phi}\,\hat{\phi}, \tag{67}$$

$$\dot{\hat{\phi}} = -\sin\theta\,\dot{\phi}\,\hat{r} - \cos\theta\,\dot{\phi}\,\hat{\theta}. \tag{68}$$

(These relations may be verified by direct differentiation of the Cartesian expansions, exactly as we did for the plane-polar case.)

17.2 Velocity

From $\vec{r} = r\hat{r}$ we obtain at once

$$\vec{v} = \dot{r}\hat{r} + r\dot{\hat{r}} = \dot{r}\,\hat{r} + r\dot{\theta}\,\hat{\theta} + r\sin\theta\,\dot{\phi}\,\hat{\phi}. \tag{69}$$

17.3 Acceleration – detailed derivation

We differentiate (69) term by term:

$$\vec{a} = \frac{d}{dt}(\dot{r}\hat{r}) + \frac{d}{dt}(r\dot{\theta}\hat{\theta}) + \frac{d}{dt}(r\sin\theta\,\dot{\phi}\hat{\phi}). \tag{70}$$

1) First term:

$$\frac{d}{dt}(\dot{r}\hat{r}) = \ddot{r}\hat{r} + \dot{r}\dot{\hat{r}} = \ddot{r}\hat{r} + \dot{r}\dot{\theta}\,\hat{\theta} + \dot{r}\sin\theta\,\dot{\phi}\,\hat{\phi}. \tag{71}$$

2) Second term:

$$\begin{aligned}\frac{d}{dt}(r\dot{\theta}\hat{\theta}) &= \dot{r}\dot{\theta}\,\hat{\theta} + r\ddot{\theta}\,\hat{\theta} + r\dot{\theta}\dot{\hat{\theta}} \\ &= \dot{r}\dot{\theta}\,\hat{\theta} + r\ddot{\theta}\,\hat{\theta} + r\dot{\theta}(-\dot{\theta}\hat{r} + \cos\theta\,\dot{\phi}\,\hat{\phi}) \\ &= -r\dot{\theta}^2\,\hat{r} + (\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta} + r\dot{\theta}\cos\theta\,\dot{\phi}\,\hat{\phi}.\end{aligned} \tag{72}$$

3) Third term:

$$\begin{aligned}\frac{d}{dt}(r\sin\theta\,\dot{\phi}\hat{\phi}) &= \dot{r}\sin\theta\,\dot{\phi}\,\hat{\phi} + r\cos\theta\,\dot{\theta}\,\dot{\phi}\,\hat{\phi} + r\sin\theta\,\ddot{\phi}\,\hat{\phi} + r\sin\theta\,\dot{\phi}\,\dot{\hat{\phi}} \\ &= \dot{r}\sin\theta\,\dot{\phi}\,\hat{\phi} + r\cos\theta\,\dot{\theta}\,\dot{\phi}\,\hat{\phi} + r\sin\theta\,\ddot{\phi}\,\hat{\phi} \\ &\quad + r\sin\theta\,\dot{\phi}(-\sin\theta\,\dot{\phi}\,\hat{r} - \cos\theta\,\dot{\phi}\,\hat{\theta}) \\ &= -r\sin^2\theta\,\dot{\phi}^2\,\hat{r} - r\sin\theta\cos\theta\,\dot{\phi}^2\,\hat{\theta} \\ &\quad + (\dot{r}\sin\theta\,\dot{\phi} + r\cos\theta\,\dot{\theta}\,\dot{\phi} + r\sin\theta\,\ddot{\phi})\hat{\phi}.\end{aligned} \tag{73}$$

Collecting coefficients of $\hat{r}$, $\hat{\theta}$ and $\hat{\phi}$ we finally obtain

$$\begin{aligned}\vec{a} &= (\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\,\dot{\phi}^2)\hat{r} \\ &\quad + (r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\,\dot{\phi}^2)\hat{\theta} \\ &\quad + (r\sin\theta\,\ddot{\phi} + 2\dot{r}\sin\theta\,\dot{\phi} + 2r\cos\theta\,\dot{\theta}\,\dot{\phi})\hat{\phi}.\end{aligned} \tag{74}$$

Newton's second law in spherical coordinates is therefore the set of three scalar equations

$$\begin{aligned}F_r &= m(\ddot{r} - r\dot{\theta}^2 - r\sin^2\theta\,\dot{\phi}^2), \\ F_\theta &= m(r\ddot{\theta} + 2\dot{r}\dot{\theta} - r\sin\theta\cos\theta\,\dot{\phi}^2), \\ F_\phi &= m(r\sin\theta\,\ddot{\phi} + 2\dot{r}\sin\theta\,\dot{\phi} + 2r\cos\theta\,\dot{\theta}\,\dot{\phi}).\end{aligned} \tag{75}$$

17.4 Physical content of the extra terms

- $-r\dot{\theta}^2$ and $-r\sin^2\theta\,\dot{\phi}^2$ are centripetal contributions arising from the curvature of the coordinate surfaces. - The Coriolis-like terms $2\dot{r}\dot{\theta}$ and $2\dot{r}\sin\theta\,\dot{\phi}$ appear whenever the radial speed is nonzero. - The mixed term $2r\cos\theta\,\dot{\theta}\,\dot{\phi}$ couples the two angular velocities.

Summary of the Component Forms

System	Acceleration components	Force equations
Cartesian	$\ddot{x},\ \ddot{y},\ \ddot{z}$	$F_i = m\ddot{x}_i$
Plane polar	$\ddot{r} - r\dot{\theta}^2, \quad r\ddot{\theta} + 2\dot{r}\dot{\theta}$	$F_r = m(\ddot{r} - r\dot{\theta}^2)$ $F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$
Spherical	see Eq. (74)	see Eq. (75)

In every case the second law retains the compact vector statement $\vec{F} = m\vec{a}$; only the concrete expression of the acceleration vector changes with the choice of coordinates.

18 Newton's Second Law in Cylindrical Coordinates

For a particle of constant mass m, Newton's second law is

$$\vec{F} = m\vec{a}. \tag{76}$$

In cylindrical coordinates, the position vector is

$$\vec{r} = \rho\,\hat{\rho} + z\,\hat{z}, \tag{77}$$

where ρ, ϕ, and z are the cylindrical coordinates, and $\hat{\rho}$, $\hat{\phi}$, and $\hat{z}$ are the corresponding unit vectors. We can write them in terms of the Cartesian unit vectors:

$$\begin{aligned}
\hat{\rho} &= \cos\phi\,\hat{\imath} + \sin\phi\,\hat{\jmath}, \\
\hat{\phi} &= -\sin\phi\,\hat{\imath} + \cos\phi\,\hat{\jmath}, \\
\hat{z} &= \hat{k}.
\end{aligned} \tag{78}$$

Their time derivatives are

$$\begin{aligned}
\frac{d\hat{\rho}}{dt} &= \dot{\phi}\,\hat{\phi}, \\
\frac{d\hat{\phi}}{dt} &= -\dot{\phi}\,\hat{\rho}, \\
\frac{d\hat{z}}{dt} &= 0.
\end{aligned} \tag{79}$$

18.1 Velocity

Differentiate the position vector:

$$\begin{aligned}
\vec{v} = \frac{d\vec{r}}{dt} &= \frac{d}{dt}\left(\rho\,\hat{\rho} + z\,\hat{z}\right) \\
&= \dot{\rho}\,\hat{\rho} + \rho\,\frac{d\hat{\rho}}{dt} + \dot{z}\,\hat{z} \\
&= \dot{\rho}\,\hat{\rho} + \rho\,\dot{\phi}\,\hat{\phi} + \dot{z}\,\hat{z}.
\end{aligned} \tag{80}$$

18.2 Acceleration

Now differentiate velocity:

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}\left(\dot{\rho}\,\hat{\rho} + \rho\dot{\phi}\,\hat{\phi} + \dot{z}\,\hat{z}\right). \tag{81}$$

Expand each term carefully:

$$\vec{a} = \ddot{\rho}\,\hat{\rho} + \dot{\rho}\,\frac{d\hat{\rho}}{dt} + \frac{d}{dt}(\rho\dot{\phi})\,\hat{\phi} + \rho\dot{\phi}\,\frac{d\hat{\phi}}{dt} + \ddot{z}\,\hat{z}. \tag{82}$$

Using

$$\frac{d}{dt}(\rho\dot{\phi}) = \dot{\rho}\dot{\phi} + \rho\ddot{\phi}, \tag{83}$$

we get

$$\vec{a} = \ddot{\rho}\,\hat{\rho} + \dot{\rho}(\dot{\phi}\,\hat{\phi}) + (\dot{\rho}\dot{\phi} + \rho\ddot{\phi})\hat{\phi} + \rho\dot{\phi}(-\dot{\phi}\,\hat{\rho}) + \ddot{z}\,\hat{z}. \tag{84}$$

Now group the $\hat{\rho}$, $\hat{\phi}$, and $\hat{z}$ terms:

$$\vec{a} = \left(\ddot{\rho} - \rho\dot{\phi}^2\right)\hat{\rho} + \left(2\dot{\rho}\dot{\phi} + \rho\ddot{\phi}\right)\hat{\phi} + \ddot{z}\,\hat{z}. \tag{85}$$

18.3 Newton's second law

Write the force in cylindrical components:

$$\vec{F} = F_\rho \hat{\rho} + F_\phi \hat{\phi} + F_z \hat{z}. \tag{86}$$

Substitute the acceleration into $\vec{F} = m\vec{a}$:

$$F_\rho \hat{\rho} + F_\phi \hat{\phi} + F_z \hat{z} = m \left[\left(\ddot{\rho} - \rho \dot{\phi}^2 \right) \hat{\rho} + \left(2\dot{\rho}\dot{\phi} + \rho \ddot{\phi} \right) \hat{\phi} + \ddot{z}\, \hat{z} \right]. \tag{87}$$

Hence, comparing components, the Newton's second law in cylindrical coordinates is

$$\boxed{\begin{aligned} &F_\rho = m \left(\ddot{\rho} - \rho \dot{\phi}^2 \right), \\ &F_\phi = m \left(2\dot{\rho}\dot{\phi} + \rho \ddot{\phi} \right), \\ &F_z = m \ddot{z}. \end{aligned}} \tag{88}$$

Supplementary Information

19 Newton's Laws of Motion

Newton's three laws of motion rest on four fundamental concepts: space (or configuration), time, mass and force. We begin by making these notions precise.

19.1 Space

Every point P in three-dimensional Euclidean space can be labelled by a position vector $\vec{r}$ that specifies both the distance and the direction of P from a chosen origin O. The most convenient way to express $\vec{r}$ is through its Cartesian components along three mutually perpendicular axes:

$$\vec{r} = x\hat{\imath} + y\hat{\jmath} + z\hat{k}. \tag{89}$$

The unit vectors $\hat{\imath}$, $\hat{\jmath}$ and $\hat{k}$ remain fixed in both magnitude and direction once we have chosen the axes.

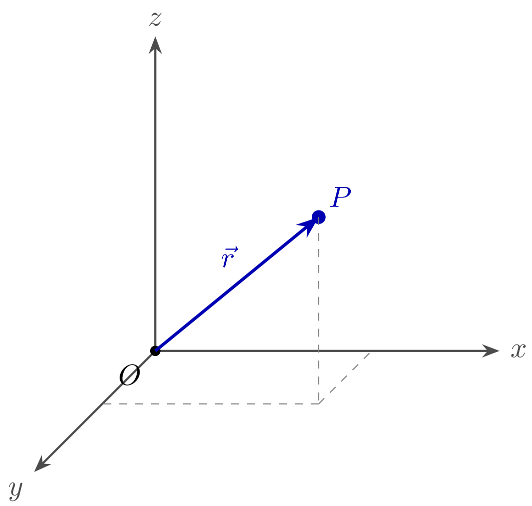

Figure 10: Position vector $\vec{r}$ of a point P relative to a fixed origin O.

19.2 Time

In classical mechanics the position of a particle is a function of a single universal time parameter t on which all observers agree:

$$\vec{r} = \vec{r}(t).$$

We define the velocity as the time derivative of the position vector,

$$\vec{v}(t) = \frac{d\vec{r}}{dt} = \lim_{\Delta t \to 0} \frac{\Delta \vec{r}}{\Delta t}, \tag{90}$$

where

$$\Delta\vec{r} = \vec{r}(t + \Delta t) - \vec{r}(t).$$

Because the Cartesian unit vectors are constant, the derivative acts component-wise:

$$\vec{v} = \frac{dx}{dt}\hat{\imath} + \frac{dy}{dt}\hat{\jmath} + \frac{dz}{dt}\hat{k}. \tag{91}$$

Each ordinary derivative is itself defined by the familiar limit

$$\frac{dx}{dt} = \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}, \qquad \Delta x = x(t + \Delta t) - x(t). \tag{92}$$

All vectors that appear in classical mechanics are assumed to be differentiable, so the limit in (90) always exists.

19.3 Reference frames

Almost every problem in classical mechanics requires the choice of a reference frame. A frame consists of

i) a spatial origin,

ii) a set of axes that label positions,

iii) a temporal origin (usually taken as $t = 0$).

A careful choice of frame can simplify the subsequent analysis. When two frames differ only by a constant spatial or temporal offset the difference is minor. A far more important distinction arises when one frame moves relative to another.

- An **inertial frame** is one in which Newton's laws take their standard form.
- A **non-inertial frame** accelerates or rotates with respect to an inertial frame; in such a frame the basic laws of motion do not hold in their simple form and fictitious forces appear.

20 Point Particles

We first state Newton's laws for a point particle (or point mass). A point particle is a convenient idealisation: an object that possesses mass but has zero size and therefore no internal degrees of freedom. It can possess translational kinetic energy, but it cannot rotate, vibrate or deform.

The laws of motion for a point particle are correspondingly simple. Later we construct the mechanics of extended bodies by regarding them as collections of many interacting point particles. In practice the idealisation is already useful: a stone thrown from a cliff, or even a planet orbiting the Sun, may be treated as a point particle for most purposes.

21 Newton's First Law (Law of Inertia)

In the absence of forces a particle moves with constant velocity $\vec{v}$ along a straight line. Equivalently, a particle remains in its state of rest or of uniform rectilinear motion unless an external force compels it to change that state. In the absence of forces the acceleration vanishes:

$$\vec{a} = \vec{0}.$$

22 Newton's Second Law

The rate of change of momentum of a particle is equal to the net force acting on it and points in the direction of that force:

$$\vec{F} = \frac{d\vec{p}}{dt}, \tag{93}$$

where the linear momentum is defined by

$$\vec{p} = m\vec{v}. \tag{94}$$

In classical mechanics the mass m is constant, so

$$\vec{F} = m\frac{d\vec{v}}{dt} = m\frac{d^2\vec{r}}{dt^2}. \tag{95}$$

Defining the acceleration

$$\vec{a} = \frac{d^2\vec{r}}{dt^2} \tag{96}$$

we recover the familiar form

$$\vec{F} = m\vec{a}. \tag{97}$$

22.1 The general problem

In the most general case the force may depend on position, velocity and time,

$$\vec{F} = \vec{F}(\vec{r}, \vec{v}, t).$$

Newton's second law then becomes a system of coupled ordinary differential equations for the trajectory $\vec{r}(t)$. An explicit closed-form solution is not always available; one therefore restricts attention to those problems that admit exact or controlled approximate solutions.

23 Newton's Third Law

The first two laws describe the response of a single particle to external forces. The third law recognises that every force on one particle is exerted by another particle. If particle 1 exerts a force $\vec{F}_{12}$ on particle 2, then particle 2 exerts an equal and opposite force $\vec{F}_{21}$ on particle 1:

$$\vec{F}_{12} = -\vec{F}_{21}. \tag{98}$$

In addition, for the great majority of forces encountered in classical mechanics the two forces act along the line that joins the two particles. Forces that possess this extra geometric property are called *central forces*. Gravity and the electrostatic interaction between point charges are familiar examples. The third law itself does not demand centrality, yet most practical forces do satisfy it.

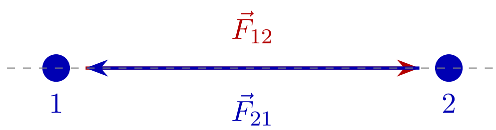

Figure 11: Action–reaction pair. The forces $\vec{F}_{12}$ and $\vec{F}_{21}$ are equal in magnitude, opposite in direction, and (for central forces) collinear with the line joining the particles.

24 Third Law and Conservation of Momentum

Consider a two-particle system. Let $\vec{F}_{12}$ be the force exerted on particle 1 by particle 2 and $\vec{F}_{21}$ the force exerted on particle 2 by particle 1. In addition each particle may experience an external force $\vec{F}_1^{\text{ext}}$ and $\vec{F}_2^{\text{ext}}$ arising from bodies outside the system. The net force on each particle is therefore

$$\begin{aligned} \vec{F}_1 &= \vec{F}_{12} + \vec{F}_1^{\text{ext}}, \\ \vec{F}_2 &= \vec{F}_{21} + \vec{F}_2^{\text{ext}}. \end{aligned} \tag{99}$$

Newton's second law applied to each particle reads

$$\begin{aligned} \dot{\vec{p}}_1 &= \vec{F}_{12} + \vec{F}_1^{\text{ext}}, \\ \dot{\vec{p}}_2 &= \vec{F}_{21} + \vec{F}_2^{\text{ext}}. \end{aligned} \tag{100}$$

The total momentum of the two-particle system is

$$\vec{P} = \vec{p}_1 + \vec{p}_2. \tag{101}$$

Differentiating with respect to time and substituting (100) yields

$$\dot{\vec{P}} = \dot{\vec{p}}_1 + \dot{\vec{p}}_2 = (\vec{F}_{12} + \vec{F}_{21}) + (\vec{F}_1^{\text{ext}} + \vec{F}_2^{\text{ext}}). \tag{102}$$

By the third law the internal forces cancel,

$$\vec{F}_{12} + \vec{F}_{21} = \vec{0},$$

and we are left with

$$\dot{\vec{P}} = \vec{F}^{\text{ext}}, \tag{103}$$

where $\vec{F}^{\text{ext}} = \vec{F}_1^{\text{ext}} + \vec{F}_2^{\text{ext}}$ is the total external force acting on the system. Equation (103) is the foundation of the mechanics of many-particle systems: as far as the total momentum is concerned, the internal forces have no effect.

If the external force vanishes,

$$\vec{F}^{\text{ext}} = \vec{0}, \tag{104}$$

then

$$\dot{\vec{P}} = \vec{0} \qquad \Rightarrow \qquad \vec{P} = \text{constant}. \tag{105}$$

Thus, in the absence of external forces the total momentum of the two-particle system is conserved.

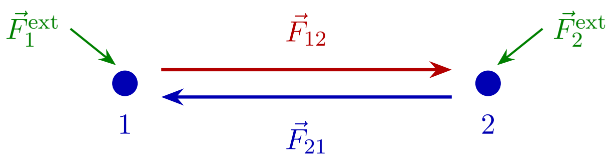

Figure 12: Two-particle system. Internal forces cancel by the third law; only the external forces change the total momentum.

25 Summary

- Space is described by a position vector $\vec{r}$; time is a universal parameter t.
- Velocity and acceleration are the first and second time derivatives of $\vec{r}$.
- Newton's first law asserts that a free particle moves with constant velocity.
- Newton's second law equates force to the rate of change of momentum, $\vec{F} = d\vec{p}/dt = m\vec{a}$ (constant mass).

- Newton's third law states that forces between two particles are equal and opposite; when the forces are central they also lie along the line of centres.
- The third law implies that internal forces cancel from the equation for total momentum, leaving $\dot{\vec{P}} = \vec{F}^{\text{ext}}$. In the absence of external forces total momentum is conserved.